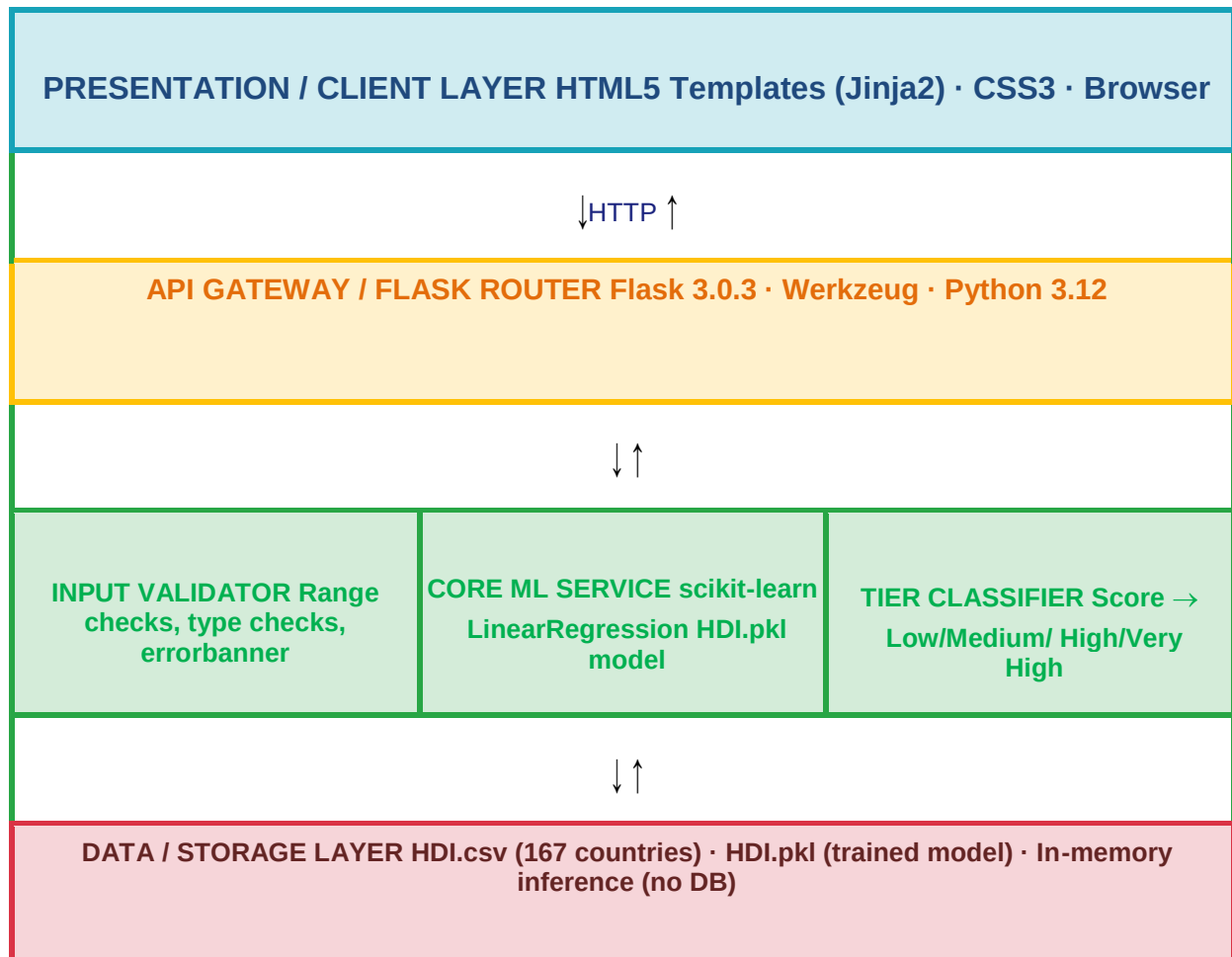


Solution Architecture

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Displays the Home, Prediction, and About pages. Collects user inputs and shows the predicted HDI score and development category.	HTML5, CSS3, Jinja2 Templates
Flask Router / API Layer	Handles GET and POST requests, loads the trained model at startup, and routes prediction requests to the ML model.	Python 3.11, Flask 3.0.3
Input Validation	Validates user inputs such as Life Expectancy, Schooling Years, and GNI per Capita to ensure valid numeric ranges before prediction.	Python, Flask
Core ML Service	Loads the trained Linear Regression model and predicts the HDI score using the input development indicators.	Scikit-learn 1.5, Pandas, NumPy, Pickle
HDI Tier Classifier	Classifies the predicted HDI score into Low, Medium, High, or Very High human development categories.	Python Functions
Data Layer	Stores the HDI dataset used for training and the serialized machine learning model used for prediction.	HDI.csv, HDI.pkl, Pandas